



Enhancing adsorbent performance for direct air capture of CO₂ by in-situ amine-grafting of layered double hydroxides

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ABSTRACT

The development of efficient and cost-effective CO₂ adsorbents is crucial for carbon capture from ultradilute conditions. Amine-grafted mesoporous materials prepared by silane chemical reactions are well-known CO₂ adsorbents with high thermal stabilities, but their capacities for direct air capture (DAC) are usually restricted by low amine loadings and CO₂ capture capacity. Herein, we present an in-situ amine grafting method that enables high amine loadings by in-situ grafting initial ultrathin LDHs nanosheets in an organic solvent. For in-situ triamine grafted Mg₂Al-CO₃ LDH (IN-TRI-LDH) amine loading is boosted to 5.91 mmol N g⁻¹ and ultimately achieving a remarkable adsorption capacity of 0.98 mmol g⁻¹ at 25 °C and 400 ppm CO₂, nearly double that of the amine-LDHs grafted through a conventional two-step process. A further 22 % enhancement of the CO₂ capacity is observed under 20 RH%. In addition, IN-TRI-LDH maintains a working capacity of 0.85 mmol g⁻¹ at a low desorption temperature of 70 °C. A 50 adsorption-desorption cyclic test under simulated humid DAC conditions shows minimal performance degradation. Such a sufficient working capacity and low desorption temperature reduces the thermal energy consumption to 2.27 GJ t⁻¹ at a desorption temperature of 40 °C.

1. Introduction

The global average atmospheric CO₂ concentration increased substantially from 285 to 419 ppm after global industrialization [1]. This increase led to rising global temperature, resulting in extreme climate disasters such as droughts and forest fires. There is therefore an urgent necessity to remove CO₂ from the atmosphere using negative emissions technologies [2]. Among them, direct air capture (DAC) by adsorption enables fast carbon capture and requires low water and land resources [3–5]. According to the International Energy Agency report, DAC technology should contribute more than 85 MtCO₂ in 2030 and around 980 MtCO₂ in 2050 to achieve net-zero emissions by 2050. However, the current capture capacity of DAC is only 0.01 MtCO₂ [6]. To accelerate the scale-up of DAC technology, the further improvement of efficient and cost-effective adsorbents is one of the key focuses. Amine-grafted DAC adsorbents form robust chemical bonds between functionalised amino groups and substrates (typically Si-O bonds formed through the condensation reaction between alkoxysilanes and hydroxyl groups).

This approach avoids physical amine stacking and mitigates the alkaline adsorption site blockage compared to polyamine impregnation, leading to excellent cyclic stability and adsorption kinetics, which are significantly important for the separation of CO₂ from the atmosphere [7].

The performance of amine-grafted DAC adsorbents is affected by aminosilanes [8–12], substrates [13–16], and grafting methods [17–21]. Among aminosilanes, the 3-[2-(2-aminoethylamino)ethylamino]propyltrimethoxysilane (TRI), a linear triamine where amine groups are separated by ethyl linkers, was preferred for amine-grafting due to its higher amine loading densities. The linear structure of TRI also reduced spatial site resistance compared to other silane amines with branched chains [12,22], which is beneficial for achieving high grafted-amine density. Considerable research has focused on TRI-grafted supports for carbon capture, particularly porous silica materials. In 2010, Sayari and coworkers first proposed grafting TRI onto mesoporous silica for CO₂ capture from air [23,24]. Subsequently, Yang and coworkers employed a series of modification strategies to increase OH group density or enhance structural characteristics of supports to enhance amine loadings,

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including boron doping for acid treatment [25], variations in the template removal method [26], hierarchical bimodal structure design [27], and pore expansion treatments [20]. However, those strategies were primarily limited to SBA-15 as a substrate material.

Other methods for adjusting amine-grafted adsorbents include modifying the grafting temperatures [21], achieving multi-layer amine grafting by introducing water into the grafting solvents [17,19,20], and changing the grafting solvents [18]. For the wet amine-grafting process, especially at a lower grafting temperature (30 °C), more water occupies the pore space of the SBA-15 and thus results in higher amine loading [21]. However, excessive water addition may lead to the interparticle polymerization of amines, causing clumping and agglomeration of SBA-15 supports. Additionally, toluene has been widely utilized as grafting solvent because its nonpolar properties are conducive to high amine loading. Nevertheless, recent research suggests that cyclohexane as a grafting solvent demonstrated superior performance in the grafting of 3-amino-propyltriethoxysilane (APS) onto SBA-15 [18].

Ordered mesoporous silicas encounter high-cost issues (e.g. SBA-15 sources from Sigma-Aldrich are currently priced at 112,000 \$/kg), and their hydrothermal stabilities, especially under steam stripping, are still problematic. Therefore, finding cost effective alternative mesoporous supports with abundant hydroxyl (OH) groups is attractive for preparing amine-grafted adsorbents with high amine loadings. Layered double hydroxides (LDHs) are a class of two-dimensional materials with remarkable tunability, the most explored sub-family adopt the generic formula $[M_{1-x}^{2+}M_x^{3+}(\text{OH})_2]^{x+}(\text{A}^{n-})_{x/n} \cdot m\text{H}_2\text{O}$ (where M is typically Mg, Zn, Cu, Mn; M' is typically Al, and Ga; A^{n-} are typically inorganic anions) [28,29]. $\text{Mg}_x\text{Al}_{1-x}\text{CO}_3$ LDHs are particularly cost-effective with commercial scale prices in the range of 2–3 \$/kg [22], annual production is kilotons at commercial facilities [30]. Inherently, individual LDH layers stack together due extensive hydrogen bonds between the surface OH groups of individual metal hydroxide laminates and facilitated by interlayer H_2O . It has been found by O'Hare and co-workers that treating the LDH aqueous wet cake with an aqueous miscible organic (AMO) solvent replaces the interlayered H_2O , allowing the efficient exfoliation/dispersion of LDH agglomerates in highly re-dispersible nanosheets upon drying [31]. AMO-LDH nanosheets feature exposed surface OH groups capable of undergoing silylation reactions with functionalised aminosilanes.

Amine-grafted AMO-LDH materials have been considered potential candidates for use as DAC adsorbents due to their excellent adsorption kinetics and cyclic stability. APS- and TRI-grafted AMO-LDH-derived supports, prepared by lab-scale co-precipitation method with acetone treatment, achieved 70 % of the total CO_2 adsorption capacity within 30 min [22]. TRI-grafted $\text{Mg}_2\text{Al-CO}_3$ LDHs, prepared by scalable fast mixing method with N, N-Dimethylformamide (DMF) treatment, displayed an average working capacity of $0.545 \text{ mmol g}^{-1}$ during 200 rapid simulated DAC cycles (each cycle lasting approximately 15 min). Under rapid mixing within 2 min in the commercial colloid mill and further DMF treatment, the well-dispersed AMO-LDH nanosheets serves as an excellent support for amine grafting [32]. However, the adsorption performance of the current amine-grafted LDHs is insufficient, and there is a notable lack of research on optimising amine grafting processes to enhance this performance.

Considering that the current amine grafting methods typically involve grafting dry LDH powders in anhydrous toluene, which may limit the capacity improvement of amine-grafted LDHs, further investigation to optimise the current grafting processes need be conducted. Previous research has demonstrated that ultrafine LDH nanosheets with high specific surface area and large pore size promote high amine loading and, consequently, adsorption capacity [32]. Therefore, optimising the grafting process while ensuring the maintenance of ultrafine LDH nanosheets is crucial. Herein, we propose a simplified in-situ amine grafting method for LDH-based adsorbents: in-situ grafting amine onto initial ultrathin LDH wet cake with organic solvent. Alongside

investigating the traditional two-step amine grafting processes: grafting amine onto LDH powders with organic solvent, the performance enhancement mechanism of in-situ amine grafting adsorbents (IN-TRI-LDH) was identified, which is achieved with minimal damage to ultrafine LDHs and maximal amine loading. The rapid adsorption and desorption characteristics and good stability of IN-TRI-LDH are experimentally demonstrated under DAC conditions. Furthermore, the advantages of IN-TRI-LDH over the amine-functionalised adsorbents are elaborated concerning the energy consumption for DAC operation. The in-situ amine grafting method offers a scalable and energy-friendly strategy to enhance adsorbent capacity under DAC conditions.

2. Experiment section

Deionised (DI) water was used in all the experimental processes. The chemical reagents, namely $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, Na_2CO_3 , NaOH, toluene, hexane, and methanol, were of analytical purity and purchased from Sinopharm Chemical Reagent. DMF (99.9 % purity) was purchased from Innochem. TRI (95 % purity) was purchased from Merck. All chemical reagents used without further purification

2.1. Synthesis of LDHs

The LDHs was synthesized using rapid mixing and nucleation in a colloid mill without further aging [33]. Taking the $\text{Mg}_2\text{Al-CO}_3$ LDHs as an example, 60 mL DI water was added into the full-back mixing colloid mill rotating at 6000 rpm. Subsequently, the solution (180 mL) containing 12 mmol $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and 6 mmol $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was added into the rapid rotating mill. Simultaneously, 90 mL alkaline solution including 9 mmol NaOH and 36 mmol Na_2CO_3 with a concentration of 0.5 mol L^{-1} was added at a constant mixing speed (6000 rpm). The rotation process of the colloid was kept for 2 min. Then the LDH precipitate was washed to neutral with DI water through centrifugal machines (denoted as M-LDH), which can be checked by the pH paper test.

2.2. Aqueous miscible organic solvent treating

The neutral LDH wet cake M-LDH was washed directly by organic solvent in order to disperse the LDH platelet agglomerates. Typically, the obtained M-LDH was washed by DMF solution (300 mL), isolated by centrifugation (8000 rpm, 3 min), and then re-dispersed in DMF solvent (300 mL) under magnetic stirring for 4 h at 25 °C. The solid was isolated and washed again by DMF solvent (300 mL) (labelled as W-LDH). The W-LDH was dried under a vacuum at 80 °C overnight, after which it was labelled as D-LDH.

2.3. Preparation of amine-grafted LDHs

For the in-situ amine grafting process, the W-LDH wet cake (0.2 g after drying) was suspended in a 15 mL DMF solution, and 0.6 mL of TRI was added dropwise at 25 °C. This dispersion was then placed in a glass tube (20 mL) subjected to an autoclave and stirred magnetically (85 °C for 16 h). The solid was also washed twice with toluene (20 mL), separated by centrifugation and dried under vacuum at 60 °C overnight. The sample prepared by the in-situ amine grafting process was denoted as IN-TRI-LDH. It is important to note that the mass of wet LDH cannot be directly measured. To obtain the corresponding wet LDH mass, 10 mL of the mixture (LDH wet cake in DMF) is removed and dried. The required mass of W-LDH is then calculated based on the concentration.

For the two-step amine grafting process, D-LDH (0.2 g) pretreated at 100 °C for 3 h in a vacuum drying oven was dispersed in DMF/toluene solution (15 mL) in a glass tube, and 0.6 mL TRI was added dropwise at 25 °C. The glass tube was placed in an autoclave and heated at 85 °C for 16 h with magnetic stirring. The sediment was washed twice with toluene, separated, and then dried under vacuum at 60 °C overnight.

Samples grafting in a toluene environment were labeled as T-TRI-LDH. Considering that the in-situ amine grafting occurs in a DMF solution while the traditional two-step amine grafting takes place in a toluene solution (Fig. 1), the prepared samples in DMF solution were also prepared to distinguish the impact of the grafting solution, which denoted as D-TRI-LDH.

Control LDH samples were prepared without amine addition. These blank samples are distinguished by the capital letter “B” in place of “TRI”. For instance, the blank sample corresponding to the in-situ amine-grafted sample IN-TRI-LDH is identified as IN-B-LDH. The resultant samples of D-TRI-LDH and T-TRI-LDH were named D-B-LDH and T-B-LDH, respectively.

2.4. Characterisation techniques

Transmission electron microscopy (TEM) coupled with an energy-dispersive spectroscopy (EDS) detector was used to characterise the morphology of the samples (TALOS F200X, Thermo Fisher, USA). The morphologies of the samples were observed using a scanning electron microscopy (SEM) apparatus (RISE-MAGNA, TESCAN, Czech). Prior to the SEM experiments, the powdered samples were sprayed with Au for 60 s. Nitrogen physisorption isotherms at 77 K were measured using an Autosorb-iQ3 (Quantachrome, USA). Prior to testing, samples were degassed for 6 h at 100 °C.

Quantitative elemental analysis of the nitrogen content was performed on a Vario EL Cube elemental analyser (ELEMENT, German). Before testing, samples were pre-treated at 100 °C under vacuum for at 1 h. The PerkinElmer (PE) TGA 8000 instrument (USA) was used to record the thermogravimetric curve. The sample was purged with a N₂ flow at 100 °C for 3 h to remove adsorbed CO₂ and H₂O. Afterwards, the sample was heated to 900 °C at a heating rate of 10 °C min⁻¹ under N₂ purging. The infrared spectra were measured using a PE spectrum 3 with MCT detector. The lattice structure of the material was analysed using an X-ray diffractometer (XRD) (Mini Flex 600, Rigaku, Japan).

²⁹Si, ¹³C and ²⁷Al MAS Solid state NMR spectra were obtained at 79.4, 100.5 and 104.2 MHz respectively (9.4 T) on a Bruker Avance IIIHD spectrometer. All samples were packed in 4 mm O.D. rotors and spun at 10 kHz. For ²⁷Al MAS Solid state NMR in order to obtain quantitative MAS spectra, a single pulse excitation was applied using a short pulse length (0.23 μs). 7000 scans were acquired with a 0.01 s delay. The ²⁷Al chemical shift is referenced to an aqueous solution of Al (NO₃)₃ (0 ppm). ¹³C CP MAS NMR spectra were measured using a cross-polarization sequence with a variable X-amplitude spin-lock pulse and spinal64 proton decoupling. 15,000 transients were acquired using a contact time of 0.5 to 1.5 ms and a recycle delay of 0.5 to 1.5 s. All ¹³C spectra were referenced to adamantane as a secondary reference. ²⁹Si CP MAS NMR spectra were measured using a cross-polarisation sequence with a variable X-amplitude spin-lock pulse and spinal64 proton decoupling. 35,000 transients were acquired using a contact time of 2 to 4 ms and a recycle delay of 1.5 s. All ²⁹Si spectra were referenced to kaolinite as a secondary reference.

A differential scanning calorimetry (DSC) apparatus (TA DSC 2500, USA) was employed to assess the specific heat capacity of the samples. The sample was compressed into a tablet (~5 mg) and treated in situ (N₂ flow at 100 °C for 40 min) to remove H₂O and CO₂ adsorbed. The temperature range was 0 °C to 150 °C, with a temperature increase rate of 10 °C min⁻¹.

The CO₂ adsorption kinetics and cyclic stability of samples were measured using a PE TGA 8000. Approximately 5 mg of samples were applied in each test. The balanced gas (N₂) keeps a constant flow rate of 60 mL min⁻¹. For adsorption tests in simulated dry air condition, the reactive gas was 1200 ppm CO₂/N₂ with a flow rate of 30 mL min⁻¹, resulting in a final concentration of 400 ppm CO₂. When tested under wet conditions, the TGA instrument was modified by adding a bubble in a thermostatic water bath at 25 °C. The reactive gas was changed to a mixture of dry 2000 ppm CO₂/N₂ gas at a flow rate of 20 mL min⁻¹ and

humid N₂ through the bubble at a flow rate of 20 mL min⁻¹. The mixture of balanced gas and reactive gas created an atmosphere of 400 ppm CO₂ and 20 % RH at 25 °C. During the cyclic adsorption-desorption process, adsorption lasted 40 min and desorption was set at 100 °C under a N₂ flow for 5 min. For the CO₂ adsorption kinetics test, samples were pre-treated at 100 °C under a N₂ flow for 120 min. The heating and cooling rates were set to 50 and 10 °C min⁻¹, respectively. The TGA unit was equipped by a water-cooled jacket to enable rapid cooling and low-temperature adsorption tests.

Breakthrough curve tests were conducted on a custom-built fixed-bed instrument (Fig. S1). The instrument mainly consisted of an adsorption column with approximately 70 mg of adsorbents, a heating furnace, a cooling oil bath, a mass flow controller supplying 400 ppm CO₂ (60 mL min⁻¹), and a CO₂ analyzer (THA100S, 0–600 ppm, ±2% full scale). Before the adsorption test, the CO₂ analyzer was calibrated with pure N₂ and a standard CO₂ concentration of 400 ppm. The adsorption column was regenerated at 100 °C with N₂ flow (100 mL min⁻¹) until the concentration of the outlet gas reached less than 1 ppm.

3. Results and discussion

3.1. Characterisation of adsorbents

Fig. 1 provides a schematic illustration of the in-situ amine grafting process on AMO-LDH used in this study compared with the more traditional two-step method. The in-situ amine grafting process involves the direct addition of the functional triamine (TRI) to the LDH DMF suspension (W-LDH), instead of using LDH powders (D-LDH) as previously employed. After heating for 16 h, and washing with toluene (to remove excess TRI and DMF), drying (to remove toluene), the adsorbent (IN-TRI-LDH) is obtained. This approach differs from the widely applied two-step amine grafting method in literature [25], which necessitates redispersing LDH powders and TRI in toluene. This modification eliminates the extra isolation and drying processes, which would be beneficial for large-scale production and overall energy saving.

As shown in TEM images (Fig. 1a), the initial aqueous LDHs (M-LDH) under the strong shear force and high-pressure friction created by the colloid mill [34] forms as finely dispersed hexagonal platelets (15–85 nm). After the AMO treatment by DMF, the hexagonal M-LDH platelets are transformed into ultrafine and curled platelets (W-LDH) with particle sizes ranging from 20 to 97 nm (Fig. 1b). Although the interlayer H₂O can be partially substituted by DMF, reducing the stacking of LDHs caused by hydrogen bonding, the LDH particles with abundant OH groups on the lamellar surface still tend to stack during the solvent removal drying process. As shown in Fig. 1c, the dried sample (D-LDH) exhibited slightly stacked nanosheets with secondary particle sizes ranging from 106 to 120 nm. Therefore, in-situ amine grafting directly on the wet LDH particles (W-LDH) can effectively avoided the stacking issue of LDHs, ensuring the maximum retention of exposed OH groups for increased amine silylation reactions and enhanced CO₂ capture capacity.

After amine grafting treatment, LDH particles stack and aggregate significantly in both in-situ and two-step amine loading processes (Fig. 1d&e). To examine solely the impact of the grafting process (85 °C in organic solvent) on LDHs, control samples without amine functionalisation were prepared. The control blank sample T-B-LDH was prepared by the two-step grafting process in toluene in the absence of the amine. The sample exhibits significant aggregation compared to its corresponding D-LDH, its secondary particle size exceeding 300 nm (Fig. S2). And a similar structural deterioration was observed when the grafting environment was changed to DMF (D-B-LDH) (Fig. S2 and Fig. S3). The reduced pores of two-step control samples are both mesopores above 10 nm, which is crucial for allowing long chain amine loading and subsequent CO₂ capture [32,35]. The in-situ control sample (IN-B-LDH) also exhibited an aggregated particle morphology. It is notable that the particle size of IN-B-LDH was less than 164 nm, a ~ 50

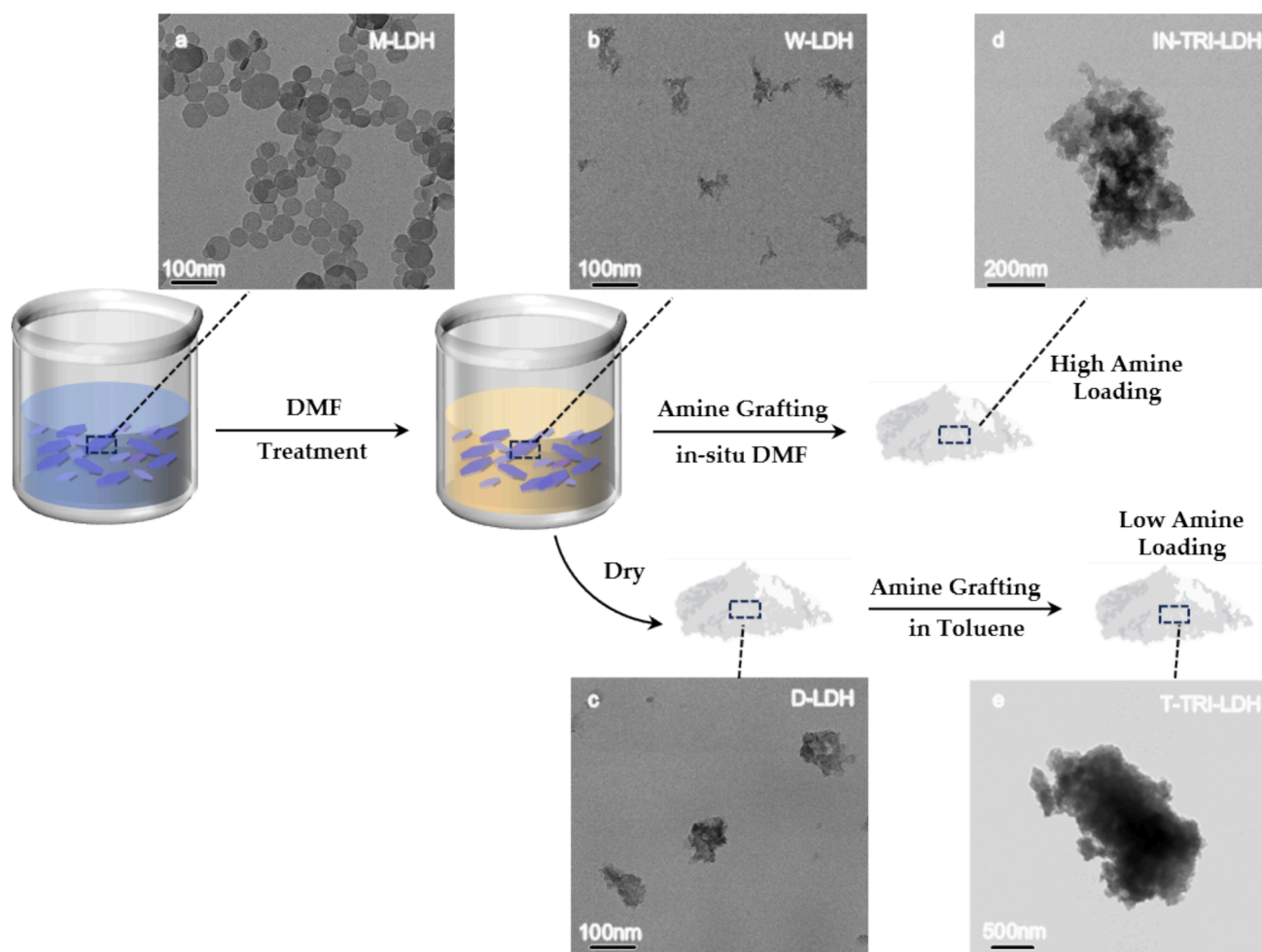


Fig. 1. Schematic of the comparison of the traditional two-step amine-grafting method (T-TRI-LDH) and in-situ amine-grafted method (IN-TRI-LDH) with TEM images of (a) the original LDHs (M-LDH), (b) the LDH wet cake treated by DMF solution (W-LDH), (c) the LDH powders treated by DMF solution (D-LDH), (d) IN-TRI-LDH, (e) T-TRI-LDH.

% reduction compared to the two-step control samples (Fig. S2). The surface area of IN-B-LDH using Brunauer-Emmett-Teller (BET) N_2 adsorption isotherms was the highest ($390.3 \text{ m}^2 \text{ g}^{-1}$) among the control samples, which indicated the in-situ grafting maintains a high concentration of available surface OH groups for the amine functionalisation process. There was no significant advantage in maintaining pore volume for the in-situ grafting process (Fig. 2a).

In conclusion, the change in grafting solvent from toluene to DMF did not account for the reduced aggregation of LDHs (T-B-LDH vs. D-B-LDH). By reducing the negative effects of drying involved in the two-step amine grafting method (W-LDH vs. D-LDH), the in-situ amine grafting method was demonstrated to be an effective approach for retaining LDH surface OH groups. More efficient OH groups are key to improving the carbon capture capacity of the amine-grafted adsorbents.

Further investigation of amine-grafted LDHs revealed that all samples exhibited a further decline in both specific surface area and pore volume in the presence of triamine TRI (Fig. 2a and Fig. 2b), which is probably unavoidable due to surface interactions and pore filling. It is interesting to note that IN-TRI-LDH retained the maximum pore volume ($0.89 \text{ cm}^3 \text{ g}^{-1}$) compared to the control samples, which is beneficial for CO_2 accessibility and uptake [20,35]. This may be attributed to the grafted long-chain TRI molecules consuming the surface OH groups and stabilising the LDH nanosheets prior to drying. The amine loading density can be determined by the nitrogen content of the samples. The in situ amine grafted adsorbent IN-TRI-LDH exhibited the highest amine grafting density of $5.91 \text{ mmol N g}^{-1}$, representing an increase of 62.8 %

compared to traditional two-step amine grafted adsorbent D-TRI-LDH (Fig. 2c). Specially, DMF used as the grafting solution (D-TRI-LDH) resulted in 19.8 % higher amine loadings on the LDH substrate compared to toluene (T-TRI-LDH). This finding is inconsistent with the existing correlation between solution polarity and amine loading amounts based on SAB-15 [18,36,37]. One potential explanation is that OH groups distribution of LDHs (surface and interlayer) is different from that of SBA-15 (porous inner wall). Another one is that the grafting base (D-LDH) has been modified by DMF, leading D-LDH particles retaining their original dispersed character in the same solvent, which is beneficial for amine grafting. Additionally, the amine loading increased with increasing grafting temperature from 30 to 100°C as shown in Fig. 2d. The amine loading reached a maximum of $6.11 \text{ mmol N g}^{-1}$ at a grafting temperature of 100°C . Higher temperature surface modifications may result in LDH aggregation, which would consequently reduce the capture capacity of LDH-based adsorbents. Further investigation is required to determine the optimal grafting temperature, which is not the focus of this study.

Further insights on the amine grafting process were investigated by comparing D-TRI-LDH and IN-TRI-LDH. D-LDH samples were employed to assess LDHs before and after grafting, given that the substrate for in-situ amine grafting is LDH wet cake. Unlike amine-impregnated adsorbents [38], there was no obvious morphological change in the SEM images (Fig. S4) after amine loading, which accounts for the lower content and uniform grafted-amine distribution. The uniform dispersion of N atoms on the LDH surface was confirmed by EDS element mapping

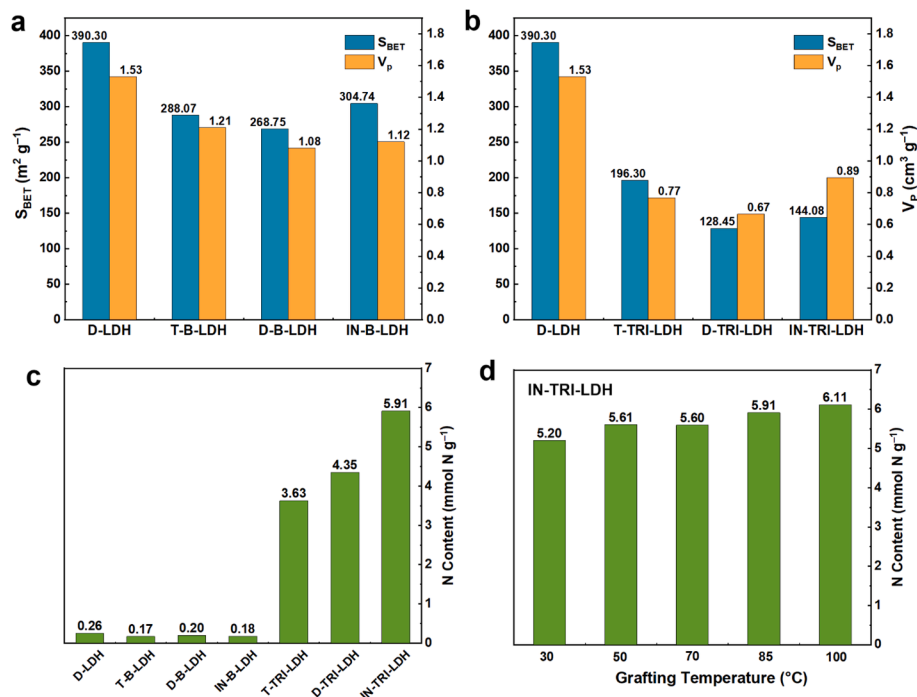


Fig. 2. The N_2 BET surface area and pore volume of D-LDH and (a) blank samples: T-B-LDH, D-B-LDH, and IN-B-LDH, (b) amine-grafted samples: T-TRI-LDH, D-TRI-LDH, and IN-TRI-LDH. Elemental analysis results of N content for (c) corresponding samples in Fig. 2a and 2b (grafting temperature: 85 °C), (d) IN-TRI-LDH grafted amine in different temperatures.

(Fig. S5). The XRD patterns of D-TRI-LDH and IN-TRI-LDH exhibited characteristic Bragg reflections for D-LDH, indicating that the structural change resulting from the two amine grafting methods was minor and similar (Fig. 3a).

The thermal stability of the materials was investigated as shown in

Fig. 3b. After pre-treating at 100 °C under an N_2 flow to completely remove adsorbed H_2O and CO_2 , all samples exhibited two characteristic weight loss events. The first weight loss at ~ 200 °C can be attributed to the loss of interlayered H_2O and/or DMF. The second weight loss at 348 °C of D-LDH is due to the removal of interlayer carbonate anions and

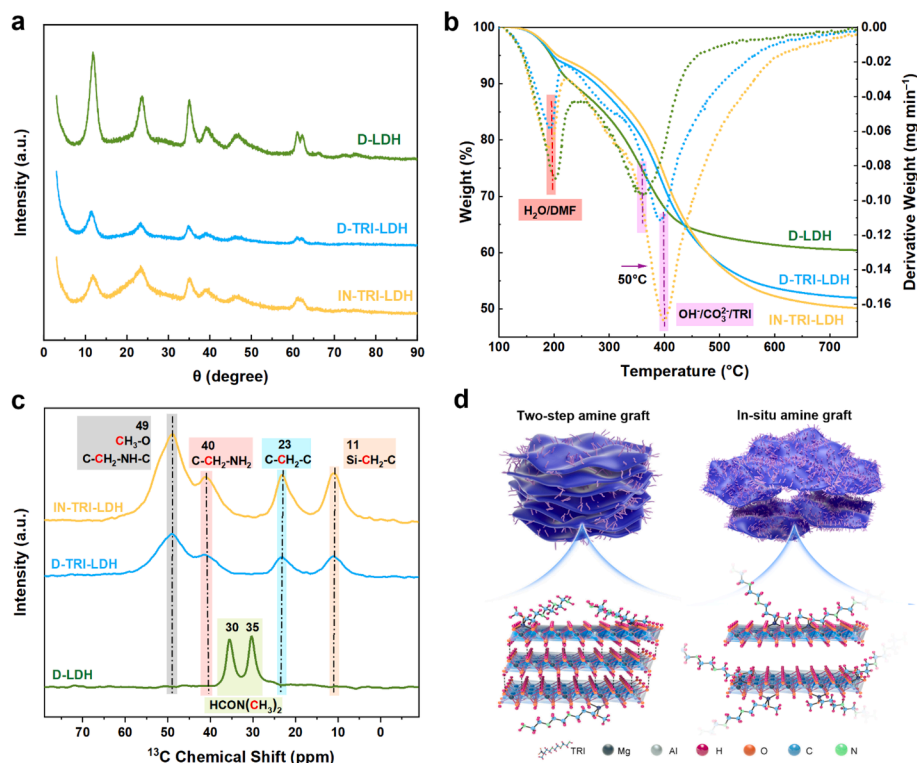


Fig. 3. (a) XRD patterns, (b) TGA analysis, (c) solid-state ^{13}C CP MAS NMR spectra of D-LDH, D-TRI-LDH and IN-TRI-LDH. (d) Comparison of traditional two-step and in-situ amine grafting methods.

OH groups from the metal hydroxide layers [39]. The second weight loss for the amine-functionalised LDHs is shifted by 50 °C to a higher temperature *ca.* 400 °C (Fig. 3b), whereas the reported polyamine-functionalised adsorbents displayed the loss of amine volatiles at temperatures between 300 and 370 °C [38,40]. This observation indicates the presence of strong interactions between TRI and the LDH supports, as well as high thermal stability. The final weight loss of IN-TRI-LDH was higher than that of D-TRI-LDH, which also proved that in situ amine grafting achieved higher amine loading. This finding agrees with the results of the N-element analysis (Fig. 2c).

The solid-state ^{13}C CP-MAS NMR spectra for the D-LDH, D-TRI-LDH and IN-TRI-LDH samples were shown in Fig. 3c. The D-TRI-LDH and IN-TRI-LDH samples each exhibited four resonances at 11, 23, 40, and 49 ppm, which are assigned as (Si-CH₂-C), (C-CH₂-C), (C-CH₂-NH₂), and (CH₃-O and/or C-CH₂-NH-C) chemical linkages in triamine TRI molecules, respectively. The observed resonances are similar to those reported for TRI-grafted SBA-15 (11, 24, 39, and 51 ppm) [41]. The resonances 35 and 30 ppm can be assigned to chemically inequivalent (N-CH₃) groups from surface bound DMF. The resonances at 35 and 30 ppm of D-LDH disappear after amine grafting. It is evidence that DMF is removed during the functionalization process [42]. The ^{13}C NMR data support the conclusion that the TRI amine was grafted onto the LDH surface without chemical modification by either the in-situ or two-step amine grafting method. The new resonance at 53 ppm for amine-grafted LDHs is observed in the solid-state ^{27}Al DPMAS spectra (Fig. S6). This may be assigned to an additional Al environment (adjacent to Si) in the amine containing samples. Both D-TRI-LDH and IN-TRI-LDH exhibited similar ^{29}Si CP MAS NMR spectra, indicating that the Si environment is the same for the two amine grafting procedures.

This modified in-situ amine grafting method (Fig. 3d), maximises and conserves active LDH nanosheet surface area, and exposed surface OH groups, facilitating higher amine grafting densities on the surface. Furthermore, the grafted amine can support the lamellar structure of LDHs after drying, resulting in a high specific surface area, pore volume and amine loading of IN-TRI-LDH, which is favourable for an improved CO₂ capture capacity.

3.2. CO₂ adsorption studies

The CO₂ adsorption performance was investigated under both simulated dry and humid DAC conditions (25 °C, 400 ppm CO₂/N₂). As illustrated in Fig. 4a, the adsorption capacities of two-step amine grafting adsorbents T-TRI-LDH and D-TRI-LDH reached 0.51 and 0.71 mmol g⁻¹, respectively. Using DMF as the grafting solution (D-TRI-LDH) resulted in a 39.2 % increase in CO₂ capacity compared to toluene (T-TRI-LDH), which can be mainly attributed to the higher amine loading (4.35 vs 3.63 mmol N g⁻¹). IN-TRI-LDH showed the highest CO₂ capacity of 0.98 mmol g⁻¹, which is nearly double that of the T-TRI-LDH.

The amine efficiency of IN-TRI-LDH (16.6 %) was 18.6 % higher than that of T-TRI-LDH (14.0 %).

The presence of moisture significantly influences the CO₂ capture performance of amine-functionalised adsorbents at ultradilute conditions [43]. To eliminate the mass gain due to moisture, the adsorbents were pre-balanced in pure N₂ with the same relative humidity (RH) under humid conditions (Fig. S7a). IN-TRI-LDH showed a 22 % increase in CO₂ adsorption capacity (1.16 mmol g⁻¹) in a humid (RH = 20 %) gas mixture of 400 ppm CO₂ in N₂ at 25 °C (Fig. S7b). When compared to other reported amine-grafted adsorbents, IN-TRI-LDH has higher amine loadings and efficiencies in both dry and humid DAC conditions (Fig. 4b). We note that achieving absolute adsorption capacities exceeding 1 mmol g⁻¹ under dry conditions and 1.2 mmol g⁻¹ under wet conditions remains a challenge for amine-grafted adsorbents. For comparison, polyamine-impregnated adsorbents, which have higher polyamine loading, can readily achieve adsorption capacities exceeding 2 mmol g⁻¹. This difference arises because the CO₂-amine reaction mechanism under anhydrous conditions requires two amine groups to react with one CO₂ molecule, so fewer amine sites in amine-grafted adsorbents lead to lower adsorption capacities [44,45].

Consequently, it is of significant importance to improve the adsorption kinetics of amine-grafted adsorbents, rather than focusing on the absolute adsorption amounts. IN-TRI-LDH reached an average adsorption rate of 1.26 mmol g⁻¹ h⁻¹ in the first 30 min. This promising CO₂ capacity and kinetics for IN-TRI-LDH under DAC condition can also be verified through the breakthrough test (Fig. S8), where the breakthrough time was 27 min, and the adsorption equilibrium time (outlet CO₂ concentration up to 95 % of the inlet) was 125 min, which is one of the highest values among the existing grafted materials (summarised in Table S1).

The performance of IN-TRI-LDH as a function of temperature was also investigated. While keeping a constant desorption temperature of 100 °C, IN-TRI-LDH captured 1.06 to 0.78 mmol g⁻¹ of CO₂ as the temperature increased from 5 to 45 °C (Fig. 5a). A low adsorption temperature is favourable for enhancing both the adsorption capacity and kinetics (Table 1). Normally, low adsorption temperatures result in low adsorption kinetics, due to the increasing CO₂ diffusion barrier caused by the decreasing temperature [46]. This opposite trend for kinetics can be attributed to the sufficient reactive gas flow (100 ml min⁻¹), which effectively covers the effect of the weakened diffusion even at low temperatures.

The desorption performance was investigated using the temperature-programmed desorption (TPD). The desorption peak for IN-TRI-LDH occurred at a low temperature of 70 °C, much lower than that of polyamine-impregnated SBA-15 (50 wt% PEI-SBA-15) (Fig. 5b). As shown in Fig. 5c, the CO₂ uptake for IN-TRI-LDH reached 0.85 mmol g⁻¹ after desorption at 70 °C, and can remain 0.74 mmol g⁻¹ after desorption at 50 °C. This may indicate that the CO₂ adsorption state of TRI-

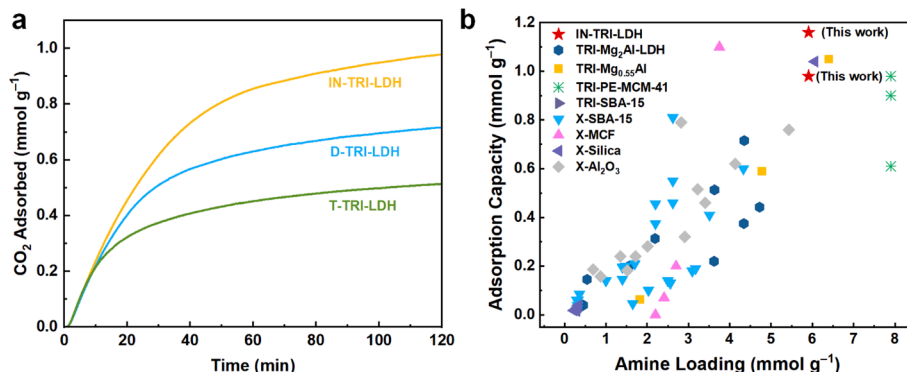


Fig. 4. (a) CO₂ adsorption kinetics of samples (desorption in N₂ at 100 °C for 2 h and adsorption in simulated dry air at 25 °C for 2 h). (b) CO₂ adsorption capacities under 0.4 mbar and 25 °C as a function of amine loadings are listed in Table S1.

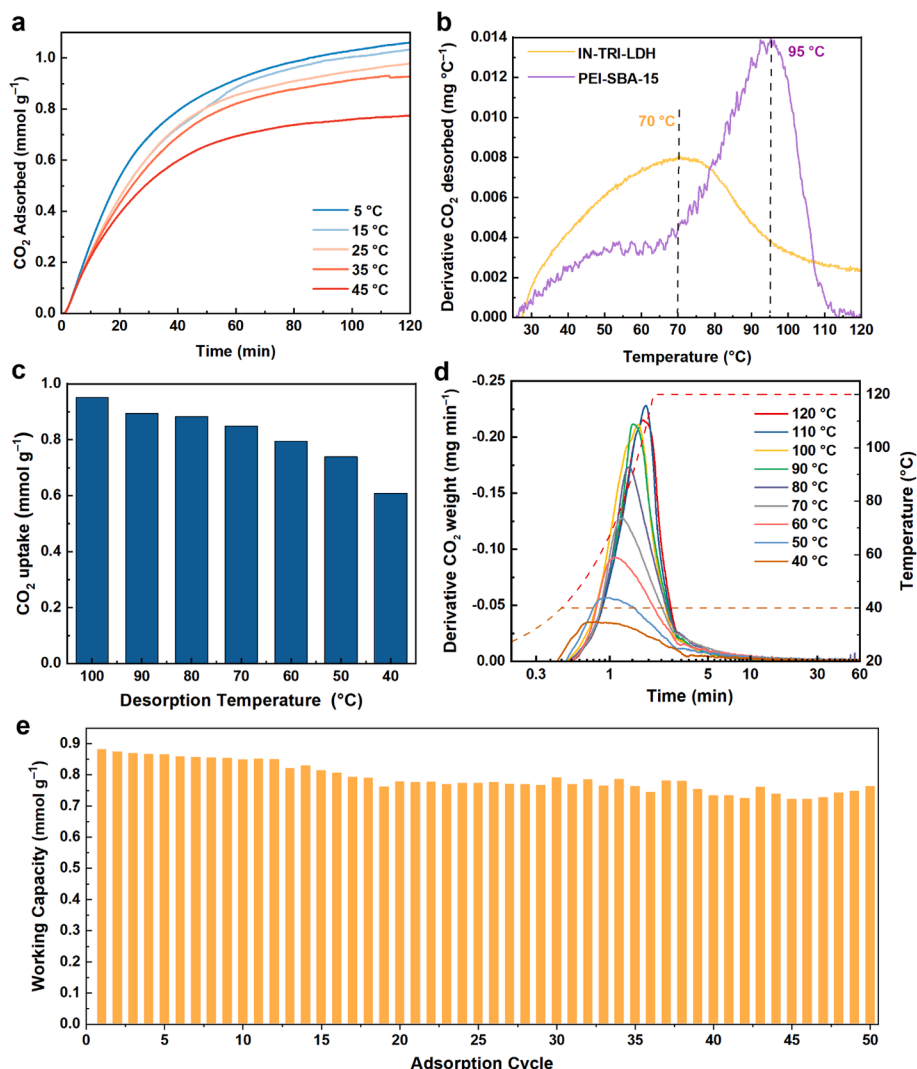


Fig. 5. The performance of IN-TRI-LDH under DAC condition (a) Adsorption capacity under different temperatures (desorption in N_2 at 100 °C for 2 h). (b) TPD results of samples at a heating rate of 5 °C min⁻¹ under N_2 purging. (c) Adsorption performance after treatment with different desorption temperatures. (d) Desorption capacity under different temperatures (adsorption in 400 ppm CO_2 balanced in N_2 at 25 °C for 2 h). (e) Stability performance over 50 cycles under simulated air conditions. Adsorption conditions: humid 400 ppm CO_2/N_2 for 40 min at 25 °C; desorption conditions: humid N_2 at 100 °C for 5 min (RH = 20 % @ 25 °C).

Table 1

The adsorption capacity and kinetics of IN-TRI-LDH under different adsorption temperature.

Adsorption Temp (°C)	Adsorption capacity (mol kg ⁻¹)	Kinetics (mol kg ⁻¹ h ⁻¹)
5	1.06	0.91
15	1.03	0.89
25	0.98	0.85
35	0.93	0.82
45	0.78	0.69

grafted LDHs for CO_2 adsorption is relatively unstable. Further investigation into the desorption temperature revealed that IN-TRI-LDH can achieve desorption within 15 min at desorption temperatures of 40–120 °C (Fig. 5d). Therefore, IN-TRI-LDH can achieve low-temperature and rapid desorption, which can significantly reduce the energy consumption for regeneration. Additionally, it is worth noting that this low-temperature desorption process (not exceeding 70 °C) eliminates oxidative degradation of adsorbents [47,48].

In addition to the single adsorption/desorption capacities, the cyclic stability of IN-TRI-LDH was characterized by TGA (Fig. 5e and

Figure S9). The rapid working cycles were conducted with desorption for 5 min (100 °C, 0.63 % H_2O , balanced N_2) and adsorption for 40 min at simulated air condition (25 °C, 400 ppm CO_2 , 0.63 % H_2O , balanced N_2). IN-TRI-LDH demonstrated a high average working capacity of 0.88 mmol g⁻¹ (per working cycle less than 1 h). Furthermore, it exhibited high hydrothermal stability after repeated activation at 100 °C in the presence of H_2O , with a CO_2 uptake capacity loss of 13 % over 50 cycles. The stability and working capacity of IN-TRI-LDH are more competitive than top performing amine-functionalised porous supports (Table S1) [49,50]. Therefore, IN-TRI-LDH could be a suitable candidate as a large-scale DAC adsorbent due to its rapid and high capture performance.

3.3. Energy consumption analysis of adsorbents

The energy consumption of DAC adsorbents remained a crucial consideration for any practical application. The thermal energy required for adsorbent regeneration (Q_{re}) accounts for approximately 80 % of the total DAC energy consumption [51]. As listed in Eq. (1), which does not consider water desorption, the Q_{re} depends on the CO_2 adsorption capacity (q_{CO_2}), the specific heat capacity of the adsorbent (C_p), adsorption-desorption temperature difference ($T_{de}-T_{ad}$) and adsorption heat

ΔH . The detailed equations and related parameters considering H_2O regeneration are given in Eq. S1-S3 and Table S2.

$$Q_{re} = \left(\frac{1}{q_{CO_2}} C_p + MW_{CO_2} C_{p,CO_2,g} \right) (T_{de} - T_{ad}) + \Delta H_{CO_2} \quad (1)$$

According to Eq. (1), high CO_2 adsorption capacity is not the sole criterion for selecting DAC adsorbents. The specific heat capacity of IN-TRI-LDH ($1.37 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-1}$) was lower than that of TRI-LDH ($1.44 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-1}$) (Fig. S10), indicating that high amine loading boosted by in-situ amine grafting method contributed to the reduction of specific heat capacity and, consequently, to the decrease in energy consumption for the capturing process. The heat of CO_2 adsorption (ΔH_{CO_2}) for IN-TRI-LDH was calculated according to adsorption isotherms at 25, 45, and 65°C (Fig. S11 and Eq. S4). The average ΔH_{CO_2} was $61.24 \text{ kJ mol}^{-1}$ for CO_2 uptakes ranging from 0.1 to 1.0 mmol g^{-1} , which is consistent with adsorbents dominated by chemisorption through amine functional groups. The chemisorption process of IN-TRI-LDH monitored by in-situ FT-IR spectroscopy, which revealed the formation of new surface bound species such as bound carbamate ($1715\text{--}1710 \text{ cm}^{-1}$ and $1520\text{--}1510 \text{ cm}^{-1}$), carbamate ions ($\sim 1430 \text{ cm}^{-1}$) during CO_2 adsorption (Fig. S12) [52]. In addition, the heat of water adsorption (ΔH_{H_2O}) for IN-TRI-LDH is $36.96 \text{ kJ mol}^{-1}$, typical for physisorption (Fig. S13) [16].

Fig. 6a illustrates the theoretical correlation between thermal regeneration energy consumption and captured CO_2 (dashed line) (Eq. S1-S3), by applying the actual variable values of corresponding adsorbents. Further enhancement of the CO_2 working capacity (from $0.8 \text{ mmol g}^{-1} \text{ h}^{-1}$) had only a minor impact on reducing energy consumption for IN-TRI-LDH. This is due to the inverse relationship between the adsorption capacity and the heat of regeneration. The energy consumption figures for IN-TRI-LDH under dry and wet conditions are marked. Despite the CO_2 adsorption capacity increasing under humid air conditions, the total energy demand for regeneration increased from 4.20 to 5.67 GJ t^{-1} due to the additional energy consumption for H_2O desorption. For comparison, three representative DAC adsorbents were selected, including polyamine-impregnated adsorbent (PEI67/Mg0.55Al-O [38,53]) and commercial adsorbents (Lewatit VP OC 1065 [54] and APDES-NFC [16,55]). The applied capacities of the Lewatit VP OC 1065 and APDES-NFC are presented without consideration of adsorption/desorption kinetics. Consequently, the actual energy cost would be higher than postulated. Table S3 presented their energy consumption under the same adsorption-desorption temperature and humid DAC conditions. IN-TRI-LDH exhibited comparable energy consumption to APDES-NFC, and lower energy consumption than PEI67/Mg0.55Al-O (because of lower H_2O uptake) and Lewatit VP OC 1065 (because of lower CO_2 adsorption heat). The above results indicate the energy consumption of adsorbents is influenced by many variables: CO_2/H_2O adsorption capacity/heat. The development of low-energy

adsorbents should not be constrained to the enhancement of CO_2 adsorption capacity alone. Furthermore, the improvement of hydrophobicity and the reduction of adsorption heat should also be considered.

Additionally, IN-TRI-LDH has the potential to further decrease the energy consumption by reducing the temperature difference between adsorption and desorption as shown in Fig. 6b. Reducing the adsorption temperature while maintaining the desorption temperature at 100°C enhanced the CO_2 adsorption capacity, but in the meantime, the increase in temperature difference resulted in a higher sensible heat for regeneration. Consequently, the regeneration energy consumption of IN-TRI-LDH initially decreased and then increased as the adsorption temperature increased from 5 to 45°C , and achieved the lowest value of 3.91 GJ t^{-1} at 35°C . In contrast, while reducing the desorption temperature from 100 to 40°C decreases the CO_2 adsorption capacity, leading to a theoretical increase in overall energy consumption, the good CO_2 capture performance of IN-TRI-LDH at lower desorption temperatures results in a net reduction in overall energy consumption (Eq. (1)). The minimum energy consumption (2.27 GJ t^{-1}) was attained at a desorption temperature of 40°C because the reduced sensible heat from 2.81 ($T_{de} = 100^\circ\text{C}$) to 0.88 GJ t^{-1} . Thus, developing adsorbents capable of desorbing at low temperatures while ensuring efficient adsorption is essential to address the high energy consumption for DAC systems.

4. Conclusion

A new improved in-situ amine grafting method has been developed for LDHs based DAC adsorbents. Compared with the traditional two-step amine grafting processes (grafting amine onto LDH powders), in-situ amine grafting method (grafting amine onto initial ultrathin LDH wet cake) eliminates the extra isolation and drying processes, facilitating the scalability and energy saving for adsorbent production. Furthermore, the process enhances the CO_2 adsorbent performance by minimising the aggregation of the LDH nanosheets and increasing the exposed OH groups for silylation reactions with the triamine (TRI). The in-situ amine grafting method provides a strategy to enhance amine loading and CO_2 capture capacity for adsorbents.

IN-TRI-LDH demonstrates both a high amine loading of $5.91 \text{ mmol N g}^{-1}$ and a superior adsorption capacity of 0.98 mmol g^{-1} in dry DAC conditions (25°C , $400 \text{ ppm } CO_2/N_2$), nearly double that of conventional two-step amine grafting LDH-based adsorbents. Even at a low desorption temperature of 70°C , IN-TRI-LDH still maintains a working capacity of 0.85 mmol g^{-1} . The stability of IN-TRI-LDH was confirmed over 50 adsorption-desorption cycles in simulated humid DAC conditions (desorption at 100°C for 5 min and adsorption at 25°C for 40 min, $RH = 20\%$). In addition, the relative high working capacity and low desorption temperatures were suggested as the key criterions to cut energy consumption for the DAC process. The operating energy consumption of IN-TRI-LDH can reach 2.27 GJ t^{-1} under dry DAC conditions, with a

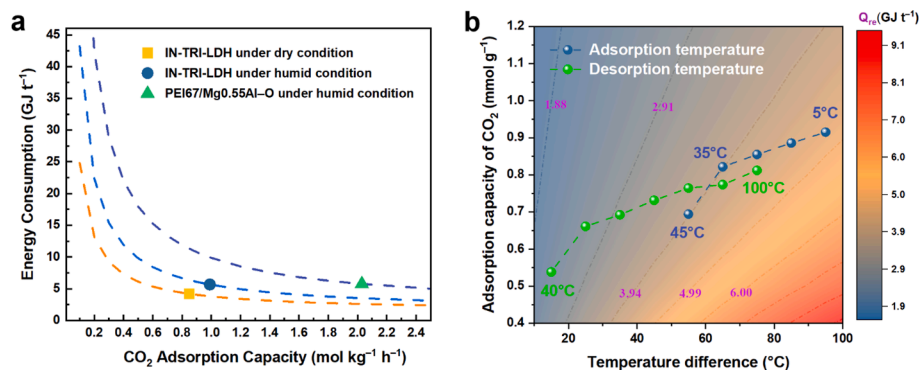


Fig. 6. The sensitivity analysis on the energy consumption of CO_2 capture: (a) H_2O and CO_2 adsorption characteristics of the adsorbent (actual values are marked). (b) Temperature difference between adsorption and desorption temperature (actual values of IN-TRI-LDH are plotted as points).

desorption temperature of 40 °C.

CRediT authorship contribution statement

Bingyao Ge: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Chunping Chen:** Writing – review & editing, Resources, Methodology. **Yifei Xu:** Investigation. **Samuel Roberts:** Writing – review & editing, Investigation. **Man Zhang:** Investigation. **Qingyang Shao:** Investigation. **Dermot O'Hare:** Writing – review & editing, Resources. **Xuancan Zhu:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xuancan Zhu reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ccej.2024.156782>.

Data availability

Data will be made available on request.

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